


```

52 |
53 | └─ T2_peptide_crystals/                ← TIER 2: HIGH d33 VALUES (7 crystals)
54 |   |
55 |   └─ dipeptides/
56 |     | └─ diphenylalanine_FF/          ← Benchmark; d33 up to 35 pC/N
57 |     |   | └─ Author_Year_FF_nanotube_selfassembly_piezo.pdf
58 |     |   | └─ Author_Year_FF_d33_PFM_measurement.pdf
59 |     |   |   └─ Yuan2023_FW_FF_coassembly_d33_35pmV.pdf
60 |     |   └─ cyclo_Phe_Phe/              ← Cyclic dipeptide
61 |     |     └─ Author_Year_cycloPP_crystal_piezo.pdf
62 |   └─ oligopeptides/
63 |     | └─ Gly_Phe_Gly/
64 |     |   └─ Author_Year_GFG_tripeptide_piezo.pdf
65 |     | └─ Phe_Phe_Gly/
66 |     |   └─ Author_Year_FFG_tripeptide_piezo.pdf
67 |     |   └─ Ala_Gly_Gly/                ← Collagen-related
68 |     |     └─ Author_Year_AGG_collagen_piezo.pdf
69 |   └─ self_assembled_peptide_crystals/
70 |     └─ Author_Year_selfassembly_peptide_piezo_review.pdf
71 |
72 |
73 | └─ T3_organic_ferroelectric/           ← TIER 3: STRONGEST PIEZO (7 crystals)
74 |   |
75 |   └─ organic_acids/
76 |     | └─ croconic_acid/               ← Proton-tautomer ferroelectric
77 |     |   └─ Author_Year_croconicAcid_ferroelectric_piezo.pdf
78 |     | └─ malonic_acid/
79 |     |   └─ Author_Year_malonicAcid_piezo_crystal.pdf
80 |     | └─ succinic_acid/
81 |     |   └─ Author_Year_succinicAcid_piezo.pdf
82 |     |   └─ tartaric_acid/              ← Historic piezo
83 |     |     └─ Author_Year_tartaricAcid_piezo_classic.pdf
84 |   └─ ammonium_molecular_ferroelectrics/
85 |     | └─ DIPAB_diisopropylammonium_bromide/ ← d33 ~23 pC/N
86 |     |   └─ Author_Year_DIPAB_ferroelectric_Science.pdf
87 |     | └─ DIPAC_diisopropylammonium_chloride/
88 |     |   └─ Author_Year_DIPAC_piezo_ferroelectric.pdf
89 |     |   └─ DIPAI_diisopropylammonium_iodide/
90 |     |     └─ Author_Year_DIPAI_piezo_crystal.pdf
91 |
92 |
93 | └─ T4_hydrogen_bonded/                 ← TIER 4: H-BOND DRIVEN (5 crystals)
94 |   |
95 |   └─ CBDC_cyclobutene_dicarboxylic_acid/
96 |     | └─ Author_Year_CBDC_Hbond_piezo.pdf
97 |     |   └─ Author_Year_CBDC.cif
98 |   └─ PhMDA_phenylmalondialdehyde/
99 |     | └─ Author_Year_PhMDA_ferroelectric_piezo.pdf
100 |     |   └─ Author_Year_PhMDA.cif
101 |   └─ croconic_acid_derivatives/
102 |     |   └─ Author_Year_croconicAcid_derivative_piezo.pdf

```

103			└─ Hbond_cocrystals/	
104			└─ Author_Year_Hbond_cocrystal_piezo.pdf	
105			└─ HFPD_hexafluoropentanediol/	← NEW: added this session
106			└─ Zhang2024_HFPD_ferroelectric_Science.pdf	← HAVE ✓
107			└─ Zhang2024_HFPD.cif	← get from supplementary
108				
109				
110			└─ T5_biomolecular/	← TIER 5: BIOLOGICAL MOLECULES (8 crystals)
111			└─ nucleobases/	
112			└─ adenine/	
113			└─ Author_Year_adenine_crystal_piezo.pdf	
114			└─ guanine/	
115			└─ Author_Year_guanine_crystal_piezo.pdf	
116			└─ cytosine/	
117			└─ Author_Year_cytosine_crystal_piezo.pdf	
118			└─ uracil/	
119			└─ Author_Year_uracil_crystal_piezo.pdf	
120			└─ thymine/	
121			└─ Author_Year_thymine_crystal_piezo.pdf	
122			└─ metabolite_crystals/	
123			└─ urea/	← Classic NLO + piezo
124			└─ Author_Year_urea_crystal_piezo_NLO.pdf	
125			└─ creatinine/	
126			└─ Author_Year_creatinine_crystal_piezo.pdf	
127			└─ uric_acid/	
128			└─ Author_Year_uricAcid_crystal_piezo.pdf	
129				
130				
131			└─ T6_sugar_based/	← TIER 6: SUGAR CRYSTALS (4 crystals)
132			└─ sucrose/	
133			└─ Author_Year_sucrose_crystal_piezo.pdf	
134			└─ glucose/	
135			└─ Author_Year_glucose_crystal_piezo.pdf	
136			└─ lactose/	
137			└─ Author_Year_lactose_crystal_piezo.pdf	
138			└─ mannitol/	
139			└─ Author_Year_mannitol_piezo_crystal.pdf	
140				
141				
142			└─ T7_amino_acid_salt/	← TIER 7: AA SALT CRYSTALS (6+ salts)
143			└─ glycine_salts/	
144			└─ glycine_lithium_nitrate_GLN/	
145			└─ Author_Year_GLN_piezo_NLO.pdf	
146			└─ glycine_phosphate/	
147			└─ Author_Year_glycinePhosphate_piezo.pdf	
148			└─ glycine_sulfate/	
149			└─ Author_Year_glycineSulfate_crystal_piezo.pdf	
150			└─ arginine_salts/	
151			└─ L_arginine_phosphate_LAP/	
152			└─ Author_Year_LAP_piezo_NLO_laser.pdf	
153			└─ histidine_salts/	

154 | | └─ L_histidine_hydrochloride/
155 | | └─ Author_Year_LHisHCl_crystal_piezo.pdf
156 | └─ alanine_salts/
157 | └─ L_alanine_acetate/
158 | └─ Author_Year_LAlanineAcetate_piezo.pdf
159 |
160 |
161 |─ T8_charge_transfer/ ─ TIER 8: ORGANIC CT CRYSTALS
162 | | ─ UPDATED: added arene_perfluoroarene
163 | |─ TTF_CA_tetrathiafulvalene_chloranil/
164 | | |─ Author_Year_TTFCA_ferroelectric_piezo.pdf
165 | | |└─ Author_Year_TTFCA_neutralionic_transition.pdf
166 | |─ TTF_derivatives/
167 | | |└─ Author_Year_TTF_derivative_ferroelectric.pdf
168 | |─ chloranil_complexes/
169 | | |└─ Author_Year_chloranil_complex_molferro.pdf
170 | └─ arene_perfluoroarene_cocrystals/ ─ NEW: added this session
171 | └─ Dunning2025_OFN2Phth_cocrystal_piezo_arXiv.pdf ─ HAVE ✓
172 |
173 |
174 |─ T9_fluorinated_molecular_crystals/ ─ NEW TIER: FLUORINATED ORGANICS
175 | | ─ Single-component fluorinated crystals
176 | | ─ with piezo from C-F dipole ordering
177 | |─ fluorinated_diols/ ─ HFPD family and analogues
178 | | |└─ (future HFPD analogues go here)
179 | |─ PVDF_related_molecular/ ─ Molecular (not polymer) fluorinated
180 | | |└─ Author_Year_fluorinated_molecular_piezo.pdf
181 | └─ perfluorinated_aromatics/ ─ OFN family single crystals
182 | └─ Author_Year_OFN_crystal_piezo.pdf
183 |
184 |
185 |─ _to_process/ ─ INBOX: drop new PDFs here first
186 | └─ (drop all new PDFs here before processing)
187
188
189

190 SECTION 2: WHAT CHANGED FROM VERSION 1.0

193 CHANGE 1 – T4 updated

194 Added: HFPD_hexafluoropentanediol/ subfolder
195 Reason: No existing T4 subfolder fit HFPD chemically
196 Status: Zhang2024_HFPD_ferroelectric_Science.pdf → HAVE
197

198 CHANGE 2 – T8 updated

199 Added: arene_perfluoroarene_cocrystals/ subfolder
200 Reason: OFN:2Phth is not classical CT crystal (no charge transfer)
201 Status: Dunning2025_OFN2Phth_cocrystal_piezo_arXiv.pdf → HAVE
202

203 CHANGE 3 – T9 NEW TIER added

204 Reason: Fluorinated molecular crystals are a growing family

205 Mechanism distinct from all other 8 tiers (C-F dipole ordering)

206 Current papers: 0 (collect next)

207

208

209

210 SECTION 3: PAPER FINDING GUIDE – TIER BY TIER

211

212

213

214 T0 – REVIEW ARTICLES

215

216 Search Terms:

217 "organic piezoelectric crystals review"

218 "molecular ferroelectric review"

219 "biomolecular piezoelectric review"

220

221 Best Sources:

222 → Chemical Reviews (ACS)

223 → Chemical Society Reviews (RSC)

224 → Nature Reviews Materials

225 → npj Asia Materials

226

227 Key Papers to Find:

228 → Horiuchi & Tokura, Nat. Mater. 2008 (organic ferroelectrics)

229 → Pan et al., Chem. Soc. Rev. 2024 (10 years molecular ferroelectrics)

230 → Guerin et al., npj Asia Mater. 2019 (organic piezo milestones)

231

232

233 T1 – AMINO ACID CRYSTALS

234

235 Search Terms:

236 "[amino acid name] crystal piezoelectric"

237 "gamma glycine piezoelectric d33"

238 "L-alanine crystal piezoresponse PFM"

239 "amino acid non-centrosymmetric crystal"

240

241 Best Sources:

242 → Google Scholar

243 → Semantic Scholar (free PDFs)

244 → PubMed (biomedical angle papers)

245 → Crystal Growth & Design (ACS)

246 → Journal of Applied Physics

247

248 Key Papers to Find:

249 → Guerin et al., Nat. Mater. 2018 (beta-glycine)

250 → Kim et al., Adv. Mater. 2020 (DL-alanine)

251 → Stapleton et al., APL 2017 (lysozyme)

252 → Yang et al., Science 2021 (gamma-glycine wafer)

253

254 Free PDF Strategy:

255 → Most on PubMed Central (PMC) free

- Semantic Scholar has author PDFs
- Search "[author name] [title] PDF" on Google

T2 – PEPTIDE CRYSTALS

Search Terms:

- "diphenylalanine piezoelectric nanotube"
- "FF dipeptide d33 piezoresponse"
- "self-assembled peptide crystal piezo"
- "cyclo dipeptide ferroelectric"

Best Sources:

- ACS Nano
- Nano Letters
- Nano Energy
- Advanced Materials

Key Papers to Find:

- Kholkin et al. (FF nanotube piezo)
- Yuan et al. 2023 (FF d33 = 35 pC/N)
- Nikiforov et al. (PFM on FF crystals)

Free PDF Strategy:

- arXiv for recent preprints
- ResearchGate (author uploaded versions)
- Unpaywall browser extension
(install at unpaywall.org – auto-finds free PDFs)

T3 – ORGANIC FERROELECTRICS

Search Terms:

- "DIPAB diisopropylammonium bromide ferroelectric"
- "croconic acid ferroelectric piezoelectric"
- "molecular ferroelectric large piezoelectric"
- "organic ferroelectric crystal d33"

Best Sources:

- Science (AAAS)
- Nature Materials
- JACS (Journal of American Chemical Society)
- Angewandte Chemie

Key Papers to Find:

- Da-Wei et al., Science 2013 (DIPAB – d33=23 pC/N)
- Horiuchi et al., Nature 2010 (croconic acid)
- You et al., Science 2017 (TMCM record piezo)

Free PDF Strategy:

- Science papers – check PubMed 1 year after publication

- JACS – check ACS AuthorChoice (some free)
- Always check: sci-hub alternatives for older papers
- IIT Delhi library access – you have institutional access

T4 – HYDROGEN BONDED CRYSTALS

Search Terms:

- "hydrogen bond ferroelectric crystal piezoelectric"
- "proton ordered crystal piezo"
- "CBDC cyclobutene dicarboxylic piezoelectric"
- "phenylmalondialdehyde ferroelectric"
- "HFPD hexafluoropentanediol piezoelectric" ← HAVE THIS

Best Sources:

- JACS
- Angewandte Chemie
- Chemistry of Materials
- Science (Zhang 2024 HFPD paper) ← HAVE THIS

Key Papers to Find:

- Horiuchi et al. (PhMDA ferroelectric)
- Katrusiak et al. (CBDC proton-ordered)
- Zhang et al., Science 2024 (HFPD) ← HAVE THIS ✓

T5 – BIOMOLECULAR CRYSTALS

Search Terms:

- "nucleobase crystal piezoelectric"
- "adenine guanine crystal piezo"
- "urea crystal piezoelectric NLO"
- "biomolecule crystal non-centrosymmetric piezo"

Best Sources:

- ACS Biomaterials Science & Engineering
- Applied Physics Letters
- Journal of Physical Chemistry
- CrystEngComm (RSC)

Key Papers to Find:

- Bhunia et al., Science 2021 (nucleobase assembly)
- Papers on urea NLO + piezo (many free on Semantic Scholar)
- Creatinine crystal structure papers (CrystEngComm)

Free PDF Strategy:

- CrystEngComm – RSC publishes some open access
 - Europe PMC for biophysics papers
 - Semantic Scholar for author manuscripts
-

T6 – SUGAR CRYSTALS

Search Terms:

"sucrose crystal piezoelectric"
"glucose crystal piezoresponse"
"sugar crystal non-centrosymmetric piezo"
"mannitol crystal piezoelectric pharmaceutical"

Best Sources:

→ Carbohydrate Research (Elsevier)
→ Crystal Growth & Design (ACS)
→ Journal of Applied Crystallography

Key Papers to Find:

→ Sucrose piezoelectric papers (P21 space group studies)
→ Mannitol polymorph piezo (pharmaceutical crystal papers)
→ Lactose crystal PFM papers

Free PDF Strategy:

→ Google Scholar → click "All versions" → find free copy
→ PubMed Central for pharmaceutical crystal papers
→ IIT Delhi library access for Elsevier journals

T7 – AMINO ACID SALT CRYSTALS

Search Terms:

"glycine lithium nitrate GLN piezoelectric NLO"
"L-arginine phosphate LAP crystal piezo"
"amino acid salt crystal ferroelectric"
"L-histidine hydrochloride crystal piezo"

Best Sources:

→ Journal of Crystal Growth (Elsevier)
→ Ferroelectrics (Taylor & Francis)
→ Spectrochimica Acta
→ Physica B

Key Papers to Find:

→ GLN papers – many from 1990s-2000s (lots of free PDFs)
→ LAP papers – laser NLO literature
→ Histidine HCl papers – Raman + piezo studies

Free PDF Strategy:

→ GLN papers are old enough to be freely available
→ Search "glycine lithium nitrate crystal site:researchgate.net"
→ Many available on ResearchGate directly
→ Ferroelectrics journal – old issues often free

T8 – CHARGE TRANSFER CRYSTALS

409

410 Search Terms:

- 411 "TTF-CA charge transfer ferroelectric piezoelectric"
- 412 "arene perfluoroarene cocrystal piezoelectric polar"
- 413 "OFN phthalazine cocrystal piezo" ← HAVE THIS
- 414 "neutral ionic transition organic ferroelectric"

415

416 Best Sources:

- 417 → Physical Review B / Letters
- 418 → JACS
- 419 → Angewandte Chemie
- 420 → arXiv cond-mat.mtrl-sci ← Dunning2025 from here ✓

421

422 Key Papers to Find:

- 423 → Oison et al., PRB 2003 (TTF-CA ab initio)
- 424 → Ferrari et al., PRB 2022 (TTF-fluoranil)
- 425 → Wiscons et al., Angew. Chem. 2018 (acenaphthene cocrystal)
- 426 → Dunning et al. 2025 (OFN:2Phth) ← HAVE THIS ✓

427

428

429 T9 – FLUORINATED MOLECULAR CRYSTALS (NEW)

430

431 Search Terms:

- 432 "fluorinated organic crystal piezoelectric"
- 433 "C-F dipole crystal ferroelectric"
- 434 "polyfluorinated molecular crystal polar"
- 435 "fluorinated diol crystal piezo"

436

437 Best Sources:

- 438 → Science / Nature (high-impact discoveries)
- 439 → JACS
- 440 → Chemistry of Materials
- 441 → arXiv cond-mat

442

443 Key Papers to Find:

- 444 → Zhang et al., Science 2024 (HFPD) ← HAVE IN T4 ✓
- 445 → Papers on PVDF analogues in molecular crystal form
- 446 → Fluorinated amino acid crystal papers

447

448 Note: This is an emerging field.

449 Not many papers yet – collect as they appear.

450

451

452

453 SECTION 4: UNIVERSAL PAPER FINDING TOOLKIT

454

455

456 FREE ACCESS TOOLS (use these first):

457

--

458 | 1. Semantic Scholar → [semanticscholar.org](https://www.semanticscholar.org) |

459 | Best for: free PDFs, citation graphs |

460			
461		2. Unpaywall Extension → unpaywall.org	
462		Best for: auto-finds legal free PDFs while browsing	
463			
464		3. PubMed Central → pubmed.ncbi.nlm.nih.gov	
465		Best for: biomedical angle papers (T1, T2, T5)	
466			
467		4. arXiv → arxiv.org/list/cond-mat	
468		Best for: physics papers (T3, T8, T9)	
469			
470		5. ResearchGate → researchgate.net	
471		Best for: author-uploaded versions of any paper	
472			
473		6. Google Scholar → scholar.google.com	
474		Best for: finding all versions of a paper	
475			
476		7. IIT Delhi Library → library.iitd.ac.in	
477		Best for: full journal access (Science, Nature, ACS)	
478			

479

480 PAID BUT ACCESSIBLE VIA IIT DELHI:

481 → Science (AAAS)

482 → Nature family journals

483 → ACS journals (JACS, Crystal Growth & Design etc.)

484 → Elsevier journals

485 → Wiley journals (Angew. Chemie, Adv. Materials)

486

487 SEARCH STRATEGY FOR EACH PAPER:

488 Step 1 → Google Scholar search

489 Step 2 → Click "All X versions"

490 Step 3 → Look for [PDF] link on right side

491 Step 4 → If no free PDF → Unpaywall extension

492 Step 5 → If still no → IIT Delhi library access

493 Step 6 → If still no → ResearchGate author request

494 Step 7 → If still no → Email corresponding author

495

496

498 SECTION 5: NAMING CONVENTION (UNCHANGED)

499

500

501 Format: Author_Year_CrystalName_JournalOrKeyword.pdf

502

503 Examples by tier:

504 T0: Horiuchi2008_organic_ferroelectrics_NatMater.pdf

505 T1: Guerin2018_betaGlycine_NatMater.pdf

506 T2: Yuan2023_FF_dipeptide_d33_35pmV.pdf

507 T3: DaWei2013_DIPAB_ferroelectric_Science.pdf

508 T4: Zhang2024_HFPD_ferroelectric_Science.pdf ← HAVE

509 T5: Author_Year_urea_crystal_NLO_piezo.pdf

510 T6: Author_Year_sucrose_P21_piezo.pdf

511 T7: Author_Year_GLN_glycineLiNitrate_piezo.pdf
512 T8: Dunning2025_OFN2Phth_cocrystal_piezo_arXiv.pdf ← HAVE
513 T9: Author_Year_fluorinated_molecular_piezo.pdf
514
515 CIF files (alongside PDF in same folder):
516 Format: Author_Year_CrystalName.cif
517 Example: Zhang2024_HFPD.cif
518 Example: Dunning2025_OFN2Phth.cif ← get from CCDC 2429146
519
520
521

522 SECTION 6: DATABASE SCHEMA (UPDATED)

525 All tiers share:

526 title, authors, DOI, year, journal, space_group,
527 d_coefficients (pC/N or pm/V), measurement_method,
528 temperature_range, crystal_system, polar_axis,
529 cif_file, csd_number, cif_source

531 Tier-specific additions:

532 T1 Amino acid → polymorph, chirality, dopant, doping_mol_pct
533 T2 Peptide → sequence, length, assembly_type, protection_group
534 T3 Ferroelec. → Ps ($\mu\text{C}/\text{cm}^2$), Ec (coercive field), Tc (Curie T)
535 T4 H-bond → H-bond_type, proton_order, switching_field
536 T5 Biomol. → molecule_type, biological_role
537 T6 Sugar → anomeric_form, polymorph, pharmaceutical_grade
538 T7 AA salt → counterion, salt_ratio, SHG_activity
539 T8 CT crystal → donor_molecule, acceptor_molecule, degree_of_CT
540 T9 Fluorinated → fluorination_degree, C_F_bond_count,
541 dipole_ordering_mechanism

545 CURRENT STATUS SUMMARY

548 PDFs Currently Collected:

549 ✓ DishonBenAmi2022_alphaGlycine_doped.pdf (T1)
550 ✓ Zhang2024_HFPD_ferroelectric_Science.pdf (T4)
551 ✓ Dunning2025_OFN2Phth_cocrystal_piezo_arXiv.pdf (T8)
552

553 CIFs Still Needed:

554 ✗ Zhang2024_HFPD.cif → get from Science supplementary
555 ✓ Dunning2025_OFN2Phth.cif → get from CCDC 2429146
556

557 Next Priority Papers to Collect:

558 Priority 1 → Guerin2018_betaGlycine_NatMater.pdf (T1)
559 Priority 2 → DaWei2013_DIPAB_Science.pdf (T3)
560 Priority 3 → Horiuchi2010_croconicAcid_Nature.pdf (T3)
561 Priority 4 → Yuan2023_FF_d33_NanoEnergy.pdf (T2)

